AP CALCULUS AB – UNIT 4

Contextual Applications of Differentiation

Expanded AP-Style Review: Full Coverage and Challenge

Name: ____________ **Class:** ____________ **Date:** ____________

Directions

This review deliberately samples every website lesson and includes a larger proportion of difficult and challenge reasoning. Part A is non-calculator. Part B is calculator-permitted. Free-response work must show setup, exact reasoning, units, and theorem conditions.

Section I – Part A: No Calculator

1. If $A(t)$ is area in square meters and t is seconds, the units of $A''(t)$ are

(A) meters per second
(B) square meters per second
(C) seconds squared per square meter
(D) square meters per second squared

2. At an instant a particle has $v < 0$ and $a > 0$. The particle is

(A) moving right and slowing down
(B) at rest
(C) moving left and slowing down
(D) moving left and speeding up

3. If $C(q)$ is cost in dollars, then $C'(500) = 7.2$ means

(A) Exactly 500 additional units cost 7.20
(B) Near 500 units, producing one additional unit costs approximately 7.20
(C) The total cost of 500 units is 7.20
(D) Cost decreases by 7.20 at 500 units

4. For a circle with changing radius, which equation correctly relates area and radius rates?

(A) $A = 2\pi r \dfrac{dr}{dt}$
(B) $\dfrac{dA}{dr} = 2\pi \dfrac{dr}{dt}$
(C) $\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$
(D) $\dfrac{dA}{dt} = \pi \left(\dfrac{dr}{dt}\right)^2$

5. If $f''(x) > 0$ near a, then the tangent-line approximation near a generally

(A) has no predictable error direction
(B) underestimates $f(x)$
(C) overestimates $f(x)$
(D) is exact for every nearby input

6. Which limit is immediately eligible for L'Hopital's Rule after substitution?

(A) $\displaystyle\lim_{x\to 0} \frac{1}{x}$
(B) $\displaystyle\lim_{x\to \infty} \frac{1}{x^2}$
(C) $\displaystyle\lim_{x\to 0} \frac{x}{x^2+1}$
(D) $\displaystyle\lim_{x\to 0} \frac{e^x - 1}{x}$

7. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars
(B) months per dollar
(C) dollars per month squared
(D) dollars per month

8. A particle has $v(t) = (t-1)^2(t-3)$. At $t = 1$, the particle

(A) has a vertical tangent
(B) is at rest but does not change direction
(C) changes from right to left
(D) changes from left to right

9. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

(A) $Q'(a) > 0$ and $Q''(a) > 0$
(B) $Q'(a) < 0$ and $Q''(a) > 0$
(C) $Q'(a) > 0$ and $Q''(a) < 0$
(D) $Q'(a) < 0$ and $Q''(a) < 0$

10. A circle's radius increases at 2 cm/s. How fast is its area increasing when $r = 5$ cm?

(A) 20π cm/s
(B) 21π cm^2/s
(C) 20π cm^2/s
(D) 10π cm^2/s

11. Use $f(x) = \sqrt{x}$ linearized at $x = 36$ to approximate $\sqrt{36 + 0.01}$.

(A) $\frac{601}{100}$
(B) $\frac{7201}{1200}$
(C) $\frac{73}{12}$
(D) $\frac{7199}{1200}$

12. The standard first step for a limit of the form 1^∞ is to

(A) differentiate the base and exponent separately
(B) replace the expression by 1
(C) apply the Product Rule
(D) take the natural logarithm and analyze the resulting product or quotient

13. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars per month
(B) dollars per month squared
(C) dollars
(D) months per dollar

14. A particle's velocity is $v(t) = (t-3)(t-6)$. On which interval does it move left?

(A) $(-\infty, 3)$
(B) $(3, 6)$
(C) $(-\infty, 3) \cup (6, \infty)$
(D) $(6, \infty)$

15. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

(A) $Q'(a) < 0$ and $Q''(a) < 0$
(B) $Q'(a) > 0$ and $Q''(a) > 0$
(C) $Q'(a) > 0$ and $Q''(a) < 0$
(D) $Q'(a) < 0$ and $Q''(a) > 0$

16. A 10-m ladder has base $x = 6$ m moving away at 0.5 m/s. Find the rate of the top.

(A) 0.375 m/s
(B) -0.375 m/s
(C) 0.5 m/s
(D) -0.5 m/s

17. Use $f(x) = \sqrt{x}$ linearized at $x = 49$ to approximate $\sqrt{49 + 0.01}$.

(A) $\frac{701}{100}$
(B) $\frac{9799}{1400}$
(C) $\frac{9801}{1400}$
(D) $\frac{99}{14}$

18. Evaluate $\lim\limits_{x \to 0} \dfrac{e^{2x} - 1}{x}$.

(A) 1
(B) 0
(C) e^2
(D) 2

19. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

(A) dollars
(B) dollars per month
(C) dollars per month squared
(D) months per dollar

20. A particle's velocity is $v(t) = (t-2)(t-4)$. On which interval does it move left?

(A) $(4, \infty)$
(B) $(2, 4)$
(C) $(-\infty, 2)$
(D) $(-\infty, 2) \cup (4, \infty)$

21. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

(A) $Q'(a) < 0$ and $Q''(a) < 0$
(B) $Q'(a) < 0$ and $Q''(a) > 0$
(C) $Q'(a) > 0$ and $Q''(a) > 0$
(D) $Q'(a) > 0$ and $Q''(a) < 0$

22. A circle's radius increases at 3 cm/s. How fast is its area increasing when $r = 3$ cm?

(A) 18π cm/s
(B) 9π cm^2/s
(C) 18π cm^2/s
(D) 19π cm^2/s

23. Given $f(2) = 6$ and $f'(2) = 4$, use linearization to approximate $f(\frac{11}{5})$.

(A) $\frac{29}{5}$
(B) 10
(C) $\frac{4}{5}$
(D) $\frac{34}{5}$

24. Evaluate $\lim\limits_{x\to\infty} \dfrac{3x^2 + 1}{e^x}$.

(A) 0
(B) The limit does not exist
(C) 3
(D) $+\infty$

Section I – Part B: Calculator Permitted

25. A particle has $s(t) = t^3 - 2t^2 + t$. Use a calculator to find $v(2.09)$.

(A) 8.54 (B) 2.483
(C) 6.744 (D) 5.744

26. A table gives $Q(3.9) = 18.42$ and $Q(4.1) = 19.18$. A centered estimate of $Q'(4)$ is

(A) 3.8 (B) 0.38
(C) 7.6 (D) 18.8

27. A sphere has radius 5.2 cm increasing at 0.18 cm/s. The volume is increasing at approximately

(A) 18.72π cm^3/s (B) 61.2π cm^3/s
(C) 19.469π cm^3/s (D) 5.616π cm^3/s

28. Using $f(2.13) \approx 0.7561$ and $f'(2.13) \approx 0.4695$, approximate $\ln(2.102)$.

(A) 0.743 (B) 0.7281
(C) 0.74 (D) 1.2256

29. Use a calculator to estimate $\dfrac{e^0.99 - 1}{0.99}$. Which value is closest?

(A) 0.99 (B) 2.708
(C) 1.708 (D) 2.691

30. At $t = 1.8$, a particle has velocity 2.7 m/s and acceleration 0.8 m/s^2. Which best describes its motion?

(A) Moving left and speeding up (B) Moving right and speeding up
(C) At rest (D) Moving right and slowing down

31. A sphere has radius $r = 2.7$ cm and $dr/dt = -0.1$ cm/s. Find dV/dt numerically.

(A) -3.393 cm^3/s (B) -8.245 cm^3/s
(C) -8.161 cm^3/s (D) -9.161 cm^3/s

32. Using $f(3.56) \approx 1.2698$ and $f'(3.56) \approx 0.2809$, approximate $\ln(3.501)$.

(A) 1.25 (B) 1.5507
(C) 1.2532 (D) 1.2108

33. Use a calculator to estimate $\dfrac{e^0.84 - 1}{0.84}$. Which value is closest?

(A) 0.84 (B) 2.316
(C) 2.567 (D) 1.567

34. A particle has $s(t) = t^3 - 2t^2 + t$. Use a calculator to find $v(1.24)$.

(A) 0.653 (B) 3.44
(C) 1.653 (D) 0.071

35. A sphere has radius $r = 4.2$ cm and $dr/dt = -0.5$ cm/s. Find dV/dt numerically.

(A) -155.17 cm^3/s (B) -110.835 cm^3/s
(C) -109.835 cm^3/s (D) -26.389 cm^3/s

36. Use a calculator to estimate $\dfrac{e^0.39 - 1}{0.39}$. Which value is closest?

(A) 2.223 (B) 1.477
(C) 0.39 (D) 1.223

Section II – Free Response

Questions 1–2 may be completed with a calculator. Questions 3–4 are non-calculator.

FRQ1 – Complete motion analysis

A particle moves on a line with velocity $v(t) = t^3 - 6t^2 + 8t$ for $0 \le t \le 6$, and $s(0) = 4$.

(a) Find all times when the particle changes direction and state the direction on each interval. [4 points]

(b) Find acceleration and determine the intervals on which speed is increasing. [4 points]

(c) Interpret $v'(3)$ in context. [2 points]

FRQ2 – Related rates with a ladder

A 10-meter ladder leans against a wall. Its foot moves away from the wall at 1.2 m/s.

(a) Find the rate at which the top moves downward when the foot is 6 m from the wall. [3 points]

(b) Find the rate of change of the angle between the ladder and the ground at the same instant. [3 points]

(c) Explain why the two negative rates have different units and meanings. [2 points]

FRQ3 – Contextual first and second derivatives

The mass $M(t)$ of a sample is measured in grams, where t is minutes. At $t = 8$, $M = 42$, $M' = -1.6$, and $M'' = 0.25$.

(a) Interpret all three values, including units. [4 points]

(b) Is the mass loss becoming faster or slower near $t = 8$? Explain. [2 points]

(c) Use a local linear model to estimate $M(8.3)$. [2 points]

FRQ4 – Linearization and propagated error

Let $f(x) = \sqrt{x}$ and use the base point $a = 25$.

(a) Find the linearization and approximate $\sqrt{25.6}$. [3 points]

(b) Determine whether the approximation is an overestimate or underestimate. [2 points]

(c) If x is measured with maximum error 0.04, estimate the maximum propagated error in f near 25. [2 points]